

GeoPandas Practice Sheet

1. Complete the hierarchy:

1. Shapely geometry
- 2.
- 3.

2. What type of object is returned by each expression? Write Shapely geometry, GeoSeries, GeoDataFrame, or pandas Series.

```
parks.geometry  
parks.geometry.iloc[0]  
parks.iloc[0]  
parks.iloc[[0]]
```

3. Explain in one sentence why `parks.geometry.iloc[0]` does not behave exactly like `parks.geometry`.

4. Write code to display all column names in `parks` as a regular Python list.

5. Write code to select only the `NAME` and `geometry` columns from `parks`.

6. Write code to select the first three rows of `parks` while keeping the result as a GeoDataFrame.

7. Write code to inspect the geometry type of every feature in `parks`.

8. What does `parks.crs` tell you?

9. Why can one point still produce a complete graph when you call `.plot()`?

10. Assume `one_park = parks.geometry.iloc[0]`. What library provides the geometry object stored in `one_park`?

```
stops.crs  
# EPSG:4326
```

11. Would EPSG:4326 be a good CRS for measuring distance in feet? Explain briefly.

12. Write code to create `stops\_nc` by reprojecting `stops` to EPSG:2264.

13. Write code to create an `area\_sqft` column in `parks\_nc` using polygon geometry area.

14. Write code to calculate the distance from every stop in `stops\_nc` to a single polygon named `park\_1`.

15. If `park\_1` is a polygon, does `.distance(park\_1)` measure distance to the polygon centroid? Explain.

16. Explain each

Predicate	Meaning
within	
contains	
intersects	
touches	

17. In `gpd.sjoin(parks, urban, predicate="within")`, which geometry is being tested as within which?

18. A park crosses an urban-area boundary. Would `within` be True? Would `intersects` be True?

19. Write code that returns how many stops are inside each polygon in a gpd named `parks` that has three polygons.

```
urban_parks = gpd.sjoin(parks, urban, predicate="within")
```

20. What is the purpose of `gpd.sjoin()`?

21. Why is a spatial join different from a regular pandas join on a common ID field?

22. Write a spatial join that keeps bus stops that fall inside park polygons. Use `stops` as the left GeoDataFrame and `parks` as the right GeoDataFrame.

23. After a point-in-polygon spatial join, how could you count how many bus stops were matched to each park? Describe the idea or write code.

24. Why can a polygon-to-polygon `intersects` spatial join create multiple output rows if each original layer was one polygon in size?

25. You want to answer: “Which parks are within 500 feet of a bus stop?” Write how would you do this with GeoPandas

- a. using a distance and apply method.
- b. using buffers